

Missing and Mixed-Up Numbers

Answer Key

Kindergarten / Grade 1 — Counting sequences and cardinality

Quick Answers

1. 7, 7

2. 13, 15

3. —

4. 8

5. 13, 14, 15, 16, 17, 15

6. —

Curriculum

Level: Kindergarten / Grade 1 — Counting sequences and cardinality

K.CC.A.1–K.CC.C.7 – *problems 1, 2, 3, 4, 5, 6*

Difficulty 1–4 of 5 across 9 tagged checks.

Common wrong answers

4. *If they got 9: jumped from 7 straight to 9, counting by twos instead of by ones*

5. *If they got 16: wrote 16 right after 14, so 15 was left out of its place*

1. Missing number in 5 6 ___ 8 9.

Solution: Start at the number just before the blank and say the next one: after 6 comes 7. Checking from the other side gives the same number, because 7 is one less than 8. 7

How to coach this: ask the child to point at each numeral and say it aloud. The blank is filled by counting, not by guessing which numeral looks right.

2. Two missing numbers in 11 12 ___ 14 ___ 16.

Solution: After 12 comes

13

and after 14 comes

15

The teen numbers are where most children slow down, because the word does not say the tens first. Counting aloud from 11 all the way to 16 is the repair, not reading the numerals silently.

3. Which is greater, 21 or 12?

Solution: 21 comes later when you count, so it is the greater number. 21 > 12

Why this one is here: 21 and 12 use the same two digits in a different order, and a child who is matching shapes rather than counting will often pick 12. Counting up out loud settles it: 12 is said long before 21.

4. Find the mistake: Ben said 6 7 9 10 11.

Solution: Ben jumped from 7 to 9. The number that belongs between them is one more than 7. 8

The count said correctly is 6, 7, 8, 9, 10, 11.

Common wrong answer: a child who is skip-counting by twos without noticing will accept 9 as the number after 7. Ask them to touch one block for each number word; the missed block is easy to see and hard to argue with.

5. Find the mistake: Mia's cards read 13 14 16 15 17.

Solution: Counting up from 13 gives 13, then 14, then 15 — but the third card says 16. The cards for 15 and 16 traded places. Put them back:

13	14	15	16	17
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The number that really follows 14 is 15

Common wrong answer: 16 right after 14. This is a misplaced number rather than a missing one — every number is present, so a child who only checks "is anything missing?" will say the row is fine. The test that works is counting the row aloud and stopping at the first word that does not match.

6. Show how you know: which number is missing in 17 18 ___ 20? (*Open response — flagged for manual review.*)

Expected answer: 19, with a drawing or a spoken explanation.

Model explanation a child might give: "I started at 18 and counted one more, and that is 19. I drew a number line and hopped one time." A drawing of 19 counters, or a number line with a hop from 18 to 19, is an equally good answer.

What to look for when you check it: the child should count forward one number at a time starting from a number before the gap, rather than guessing. The strongest sign of understanding is being able to say that the missing number is one more than the number before it — and, if you ask, one less than the number after it.